

AIM

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu Server, configure the required network parameters such as IP address, subnet mask, gateway, and DNS, and establish secure remote access through SSH.

WHAT IS VPS?

A **Virtual Private Server (VPS)** is a virtual server created using virtualization technology. It works as an independent server with its own operating system, storage, memory, and network configuration, even though the underlying physical hardware may be shared with other virtual machines.

Virtualization helps in making better use of physical computing resources and is an important concept used in cloud computing environments.

NETWORK CONFIGURATION PARAMETERS**1. IP Address**

An **IP address** is a unique numerical address assigned to a device so that it can be identified and communicated with over a network.

2. Subnet Mask

A **subnet mask** is used to identify which part of an IP address represents the network and which part represents the host.

3. Gateway

A **gateway** acts as an entry and exit point that allows a device to communicate with networks outside its local network.

4. DNS (Domain Name System)

DNS translates human-readable domain names into their corresponding IP addresses, allowing systems to locate websites and other network services.

5. SSH (Secure Shell)

SSH is a secure communication protocol that allows users to remotely access and manage a Linux-based system.

PROCEDURE**Step 1 – Check Internet Connectivity:**

Open the Ubuntu terminal and test the internet connectivity of the system by executing the command `ping google.com`.

```
student@Ubuntu-Server:~/Desktop$ ping -c 8 google.com
PING google.com (142.250.196.14) 56(84) bytes of data:
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=1 ttl=57 time=28.4 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=2 ttl=57 time=27.9 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=3 ttl=57 time=41.2 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=4 ttl=57 time=30.6 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=5 ttl=57 time=26.8 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=6 ttl=57 time=33.1 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=7 ttl=57 time=29.7 ms
64 bytes from maa05s-in-f14.1e100.net (142.250.196.14): icmp_seq=8 ttl=57 time=31.5 ms

--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7012ms
rtt min/avg/max/mdev = 26.8/31.15/41.2/4.3 ms
```

Step 2 – Check Network Routing:

Display the current network routing details of the Ubuntu system using the command `ip route`.

```
student@Ubuntu-Server:~/Desktop$ ip route
default via 192.168.56.2 dev enp0s3 proto dhcp src 192.168.56.15 metric 100
192.168.56.0/24 dev enp0s3 proto kernel scope link src 192.168.56.15 metric 100
192.168.100.0/24 dev br-ex proto kernel scope link src 192.168.100.1
```

Step 3 – Verify DNS Configuration:

Check the DNS configuration by executing the command `cat /etc/resolv.conf`.

```
student@Ubuntu-Server:~/Desktop$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved.
#
# Run "resolvectl status" to see details about the uplink DNS servers currently in use.
#

nameserver 192.168.56.2
nameserver 8.8.8.8
options edns0 trust-ad
search .
```

Step 4 – Update Package Repository:

Update the Ubuntu package repository information, if necessary, by using the command:

```
sudo apt update
```

Step 5 – Install OpenSSH Server:

Install the **OpenSSH Server** package on the Ubuntu system using:

```
sudo apt install openssh-server -y
```

Step 6 – Enable SSH Service:

Configure the SSH service to start automatically whenever the system boots by executing:

```
sudo systemctl enable ssh
```

Step 7 – Start SSH Service:

Start the SSH service using the command:

```
sudo systemctl start ssh
```

Step 8 – Check SSH Service Status:

Verify the current status of the SSH service with:

```
sudo systemctl status ssh
```

```
student@Ubuntu-Server:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 10:12:47 IST; 3min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1842 (sshd)
      Tasks: 1 (limit: 4643)
     Memory: 5.4M
        CPU: 78ms
    CGroup: /system.slice/ssh.service
            └─1842 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Step 9 – Find the IP Address:

Obtain the IP address assigned to the Ubuntu system by executing:

```
hostname -I
```

Step 10 – Establish SSH Connection:

Establish a remote SSH connection to the Ubuntu system using the following command:

```
ssh <username>@<IP_address>
```

Step 11 – Verify SSH Login:

Confirm that the SSH connection has been established successfully by checking for the “**Welcome to Ubuntu**” message displayed in the terminal.

```
student@Ubuntu-Server:~/Desktop$ ssh student@192.168.56.15
The authenticity of host '192.168.56.15 (192.168.56.15)' can't be established.
ED25519 key fingerprint is: SHA256:aQ7mZk1oWq93XvRs4jLn0bTf8yHc2eUp6NdVw9gkpQ8
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.15' (ED25519) to the list of known hosts.
student@192.168.56.15's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-45-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu 24 Sep 2026 10:16:03 AM IST

  System load:  0.08               Processes:      112
  Usage of /:   21.4% of 24.51GB    Users logged in: 1
  Memory usage: 18%               IPv4 address for enp0s3: 192.168.56.15

Last login: Thu Sep 24 09:58:11 2026 from 192.168.56.1
```

CONCLUSION

The Ubuntu Server environment was successfully configured with the required network settings and its network connectivity was verified. The **OpenSSH Server** was installed, enabled, and started successfully. The system IP address was obtained and used to establish a remote SSH connection. The successful appearance of the Ubuntu welcome message confirmed that secure remote access to the system was working correctly.